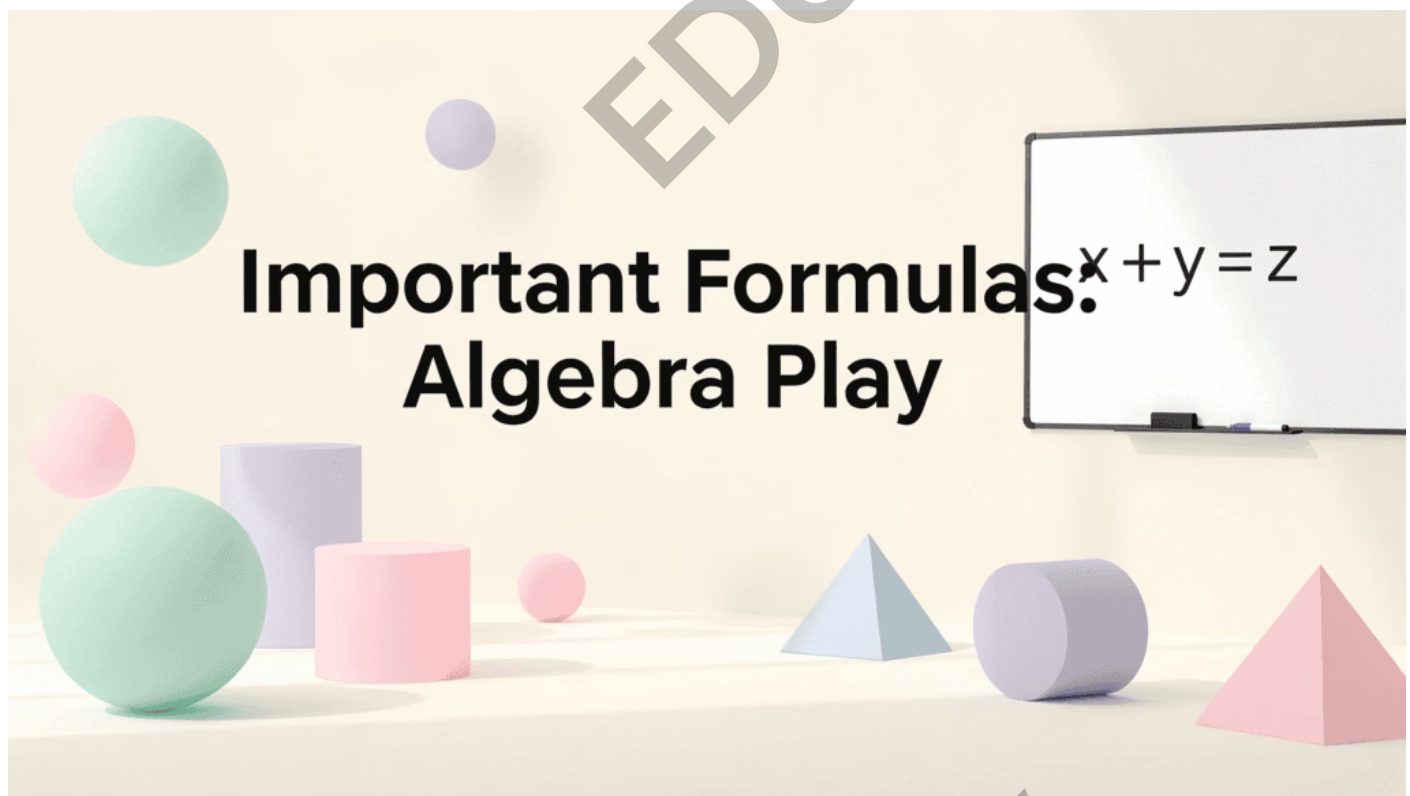




Important Formulas: Algebra Play

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Think of a Number Tricks

Date Trick Formula

Let Month = M and Day = D

Steps:

1. Multiply M by 5: $5M$

2. Add 6: $5M + 6$

3. Multiply by 4: $4(5M + 6) = 20M + 24$

4. Add 9: $20M + 24 + 9 = 20M + 33$

5. Multiply by 5: $5(20M + 33) = 100M + 165$

6. Add day D: $100M + 165 + D$

Formula to find original date:

Date = Final Answer - 165

The last 2 digits give the Day (D) and remaining digits give the Month (M).

Example: If final answer = 291

$291 - 165 = 126$

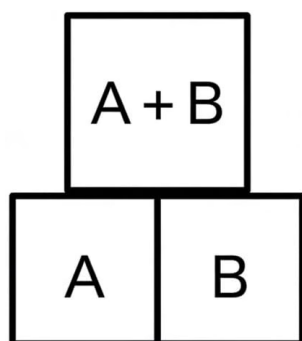
Month = 1, Day = 26 (26th January)

Number Pyramids

Basic rule for Number Pyramids

Formula for top number:

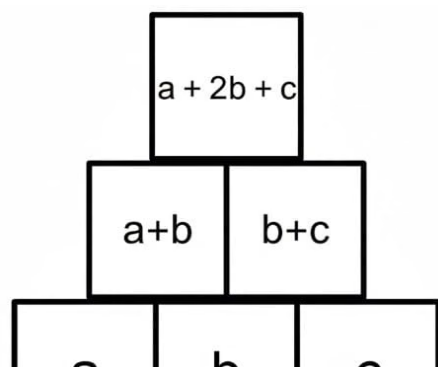
2-row pyramid:



Top = $a + b$

Formula for 3-row pyramid

3-row pyramid:



a	b	c
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Formula: $\text{Top} = a + 2b + c$

Explanation: The middle number (b) appears twice in the top calculation.

Formula for 4-row pyramid

4-row pyramid:

$a+3b+3c+d$			
$a+2b+c$		$b+2c+d$	
$a+b$	$b+c$	$c+d$	
a	b	c	d

Formula: $\text{Top} = a + 3b + 3c + d$

Pattern: Coefficients follow Pascal's triangle pattern (1, 3, 3, 1)

General formula for n-row pyramid

The coefficients of the bottom row values follow binomial coefficients.

For n rows:

$$\text{Top} = C(n-1,0) \times a_1 + C(n-1,1) \times a_2 + C(n-1,2) \times a_3 + \dots + C(n-1,n-1) \times a_n$$

where $C(n,r)$ represents binomial coefficients.

Calendar Magic & Algebra Grids

Formula for 2×2 calendar grid sum

Let top-left number = a

a	a + 1
a + 7	a + 8

Formula: $\text{Sum} = a + (a+1) + (a+7) + (a+8) = 4a + 16$

To find 'a' when sum is given:

$$a = (\text{Sum} - 16) / 4$$

Example: If sum = 40

$$4a + 16 = 40$$

$$4a = 24$$

$$a = 6$$

So the grid is:

6	7
13	14

Formula for 3×3 calendar grid

Let top-left number = a

a	a+1	a+2
a+7	a+8	a+9
a+14	a+15	a+16

Formula: Sum = $9a + 72$

To find 'a': $a = (\text{Sum} - 72) / 9$

The Largest Product

Q: How to arrange digits to form the largest product in $ab \times c$ format?

Ans: Given three digits where $p < q < r$:

Rule: Place the largest digit (r) as the multiplier, and arrange the other two digits in descending order for the multiplicand.

Formula: Largest Product = $(10q + p) \times$

Example: For digits 2, 3, 5:

$$\text{Largest product} = 32 \times 5 = 160$$

Algebraic Proof:

$$qp \times r = (10q \times r) + (p \times r)$$

$$rp \times q = (10r \times q) + (p \times q)$$

Since $r > q$, we have $(p \times r) > (p \times q)$

Therefore $qp \times r > rp \times q$



